

Alan Linguist

Diagnosis
Dedifferentiated chondrosarcomaAccession No.
TL-24-AU2DMEAY

xT

Date of Birth
06/07/1955Sex
MalePhysician
John CharlsonInstitution
MCW - Froedtert Hospital Cancer
Center
E2304021TEMPUS | xT
648 gene panelTumor specimen:
Soft tissue, chest wall
Wisconsin Diagnostics Laboratories -
Pathology 1-5-23-18596, D1
Collected 8/1/2023
Received 5/28/2024
Tumor Percentage: 80%
(post microdissection)

GENOMIC VARIANTS

Potentially Actionable

Variant Allele Fraction

CHEK2

c.846+4_846+7del Splice region variant - LOF 85.6%

Biologically Relevant

CDKN2A

Copy number loss

CDKN2B

Copy number loss

LYN

Copy number gain

MTAP

Copy number loss

Tumor / Normal Matched Analysis (Potential Germline)

No normal sample was received, therefore tumor/normal matched analysis was not performed.

IMMUNOTHERAPY MARKERS

Tumor Mutational Burden

4.7 m/MB

53rd percentile

Microsatellite Instability Status

Stable

Equivocal

High

FDA-APPROVED THERAPIES, OTHER INDICATIONS

PARP Inhibitor

Olaparib

NCCN, Consensus, Prostate Cancer

CHEK2 c.846+4_846+7del Loss-of-function

We were unable to determine whether treatments on this report were previously prescribed for this patient.

CLINICAL TRIALS

TAPUR: Testing the Use of Food and Drug Administration
(FDA) Approved Drugs That Target a Specific Abnormality in
a Tumor Gene in People With Advanced Stage Cancer
(NCT02693535)

Phase II

Wauwatosa, WI - 0 ml
✓ CHEK2 c.846+4_846+7del
mutation
✓ CDKN2A deletion
✓ CDKN2B deletion

TEMPUS

Electronically Signed By
Rongjin Ren, MD, PhDCLIA Number
14D2114007Date Signed/Reported
06/05/2024Laboratory Medical Director
Brett Mahon, MD, FCAPTempus ID #
TL-24-AU2DMEAYPipeline Version
5.1.0

1/8

Tempus AI, Inc. • 600 West Chicago Avenue, Suite 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com

CLINICAL TRIALS (CONTINUED)

Alan Linquist | TL-24-AU2DMEAY

Safety and Tolerability of TNG908 in Patients With MTAP-deleted Solid Tumors
(NCT05275478)

Phase I/II
Chicago, IL - 82 mi
✓ MTAP deletion

AMG 193, Methylothioadenosine (MTA) Cooperative Protein Arginine Methyltransferase 5 (PRMT5) Inhibitor, Alone and in Combination With Docetaxel in Advanced Methylothioadenosine Phosphorylase (MTAP)-Null Solid Tumors (NCT05094336)

Phase I/II
Hinsdale, IL - 86 mi
✓ CDKN2A deletion
✓ MTAP deletion

HUMAN LEUKOCYTE ANTIGEN (HLA) CLASS I GENOTYPING

HLA class I genotyping is determined by using DNA sequencing data and is only intended for clinical trial matching.

Gene	Allele I	Allele II
HLA-A*	A*03:01*	A*03:01*
HLA-B*	B*07:02*	B*07:02*
HLA-C*	C*07:02*	C*07:02*

* This result was produced from analysis of the tumor specimen because the normal was unavailable or did not meet quality requirements for HLA class I genotyping results.

* This is the most representative allele within a group of alleles that are identical at exons 2 and 3 and cannot be disambiguated. For a full list of group members see Tempus Document Library.

VARIANTS OF UNKNOWN SIGNIFICANCE

Gene	Mutation effect	Variant allele fraction
TPMT	c.356A>C p.K119T Missense variant NM_000367	93.1%
EPHB2	c.3067G>A p.G1023R Missense variant NM_001309193	90.4%
SETBP1	c.584G>T p.G195V Missense variant NM_001130110	89.5%
PTPN22	c.2291dup p.N764fs Frameshift NM_015967	88.1%
IRF2	c.88-4dup Splice region variant NM_002199	28.9%
TBL1XR1	c.1022C>T p.P341L Missense variant NM_024665	10.6%
PREX2	c.1628G>T p.G543V Missense variant NM_024870	10.3%
LRP1B	c.1439A>G p.Y480C Missense variant NM_018557	9.9%
CTC1	c.2759-4A>G Splice region variant NM_025099	7.8%
IDO1	c.696A>G p.I232M Missense variant NM_002164	7.0%

TEMPUS

Electronically Signed By: CLIA Number: Date Signed/Reported: Laboratory Medical Director: Tempus ID #: Pipeline Version: 2/8
 Rongjin Ren, MD, PhD 14D2114007 06/05/2024 Brett Mahon, MD, FCAp TL-24-AU2DMEAY 5.1.0
 Tempus, LLC • 500 West Chicago Avenue, Ste 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com

VARIANTS OF UNKNOWN SIGNIFICANCE (CONTINUED)

Alan Linquist | TL-24-AU2DMEAY

Gene	Mutation effect	Variant allele fraction
RP515	c.131A>G p.Y44C Missense variant NM_001308226	5.1%

LOW COVERAGE REGIONS

TAF1

VARIANT DETAILS - POTENTIALLY ACTIONABLE

CHEK2 c.846+4_846+7del NM_007194 Splice region variant - LOF

VAF: 85.6%

CHEK2 is a tumor suppressor gene involved in regulating cellular responses to DNA damage including DNA repair, cell cycle arrest, and apoptosis. CHEK2 activation leads to downstream activation of TP53. Loss of function mutations and copy number loss of CHEK2 are associated with cancer progression.

VARIANT DETAILS - BIOLOGICALLY RELEVANT

CDKN2A Copy number loss

CDKN2A encodes two proteins, p16(INK4a) and p14(ARF), that function in regulating cell growth. The p16(INK4a) protein regulates the cell cycle through the inhibition of CDK4 and CDK6, preventing them from stimulating cell proliferation. The p14(ARF) protein binds to MDM2 to keep p53 intact and stimulate the p53-dependent cell cycle arrest and apoptosis. Loss of function mutations, copy number loss, and underexpression of CDKN2A are associated with cancer progression.

CDKN2B Copy number loss

CDKN2B encodes p15(INK4b), a protein that functions in regulating cell growth. The p15(INK4b) protein regulates the cell cycle through the inhibition of CDK4 and CDK6, preventing them from stimulating cell proliferation. Loss of function mutations, copy number loss, epigenetic variation, and underexpression of CDKN2B are associated with cancer progression.

LYN Copy number gain

LYN encodes a protein tyrosine kinase in the SRC family that is involved in B-cell receptor stimulation and regulates mast cell degranulation and erythroid differentiation. Copy number gains, fusions, epigenetic variation, and overexpression of LYN are associated with cancer progression.

MTAP Copy number loss

MTAP encodes an enzyme involved in polyamine metabolism and the regeneration of methionine and adenine. MTAP is often co-deleted with CDKN2; however, it has been shown to be independently involved in regulation of cell migration and invasion. Loss of function mutations, copy number loss, epigenetic variation, and underexpression of MTAP are associated with cancer progression.

TEMPUS

Electronically Signed By: CLIA Number Date Signed/Reported Laboratory Medical Director Tempus ID # Pipeline Version
 Rongqin Ren, MD, PhD 14D2114007 06/05/2024 Brett Mahon, MD, FCAP TL-24-AU2DMEAY 5.1.0 3/8
 Tempus AI, Inc. • 600 West Chicago Avenue, Ste 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com

Alan Linquist | TL-24-AU2MEAY

Assay Description

The Tempus xT (version 4) assay is a custom oncology testing panel consisting of 648 genes with single nucleotide variants (SNVs), insertions and deletions (indels), copy number variants (CNVs), and chromosomal rearrangements (translocations) detected by hybrid capture next-generation sequencing (NGS) using custom designed IDT probes. A complete gene list can be found at the end of this assay description. This assay has 98.2% sensitivity for SNVs above 5% variant allele fraction (VAF), 91.1% sensitivity for indels above 5% VAF, and 90.9% sensitivity for reported chromosomal rearrangements. Certain driver or resistance genes may be reported to lower VAF when technically possible. The assay has 91.4% sensitivity for CNVs given the tumor purity limit of detection (LOD) for copy number gains at 30% and copy number loss at 40% tumor purity. The assay has 90.5% sensitivity for microsatellite instability (MSI) detection at 30% tumor purity and 100% sensitivity for tumor mutational burden high (TMB-high). Detection at 30% tumor purity and 100% sensitivity for tumor mutational burden high (TMB-high). Human leukocyte antigen (HLA) class I genotyping is reported from normal-matched analysis with 100% accuracy for HLA-A, 99.0% accuracy for HLA-B, and 100% accuracy for HLA-C. HLA genotyping from tumor analysis is reported with 100% accuracy for HLA-A, 97.9% accuracy for HLA-B, and 99.0% accuracy for HLA-C. Test results ordered before March 6, 2024 may not include HLA genotyping results.

Human Leukocyte Antigen class I (HLA-A, HLA-B, HLA-C) reflects the genotype status of the patient and is assessed from DNA sequencing reads. The Tempus HLA class I genotyping device analyzes reads mapping to any known IMGT HLA class I allele. HLA mapped reads are used to select candidate alleles and run the open source software HLA-IA to determine the final genotype. Reporting of an individual HLA class I allele is provided at the gene level by analysis of either the patient's normal sample or tumor sample. Reporting preference is given to the normal analysis whenever a sufficient normal sample is provided. If the tumor sample is used to identify the HLA genotype, it will be indicated on the report. Data from both the tumor and the normal analysis generally reflect the patient's germline HLA status. The full set of reported alleles may be generated from a mixture of tumor and normal analyses. HLA genotyping is reported at the 4-digit level, which identifies alleles that code for the same protein. The nomenclature used is based on IMGT/HLA version 3.4.0. HLA genotyping results may support referring patients to clinical trials based on specific genotype.

The clinical summary (first page of the report) shows actionable and biologically relevant somatic variants, and certain pathogenic or likely pathogenic variants that are reported as potential germline findings (if a matched normal sample was provided and analyzed). The information from a matched normal may include potential clonal hematopoiesis of indeterminate potential (CHIP) variants indicated in the Notes section. Detection of potential CHIP variants through the xT assay is only provided from tumor and normal matched analysis of the 50 most common CHIP-associated genes. Variants present at the same or lower VAF in the tumor compared to the normal, or variants present in the normal but not identified in the tumor, may be derived from CHIP. Potential CHIP variants are noted at the discretion of the pathologist signing the report. Reportable potential germline findings are limited to genes and variants associated with inherited cancer syndromes. **Potentially Actionable** alterations have an associated therapeutic, diagnostic, and/or prognostic indication, based on evidence from US-based clinical guidelines and medical literature curated in the Tempus knowledge database. **Biologically Relevant** alterations may have functional significance or an association with the disease state in the medical literature, but do not have relevant therapeutic, diagnostic, or prognostic evidence curated in the Tempus knowledge database. **Variables of Unknown Significance (VUS)** are alterations exhibiting an unclear effect on function and/or lack sufficient evidence to determine pathogenicity. **Pathogenic variants are not reported. Low Coverage Regions** are included when mean coverage over any region(s) of a gene falls below a threshold of 35x in the tumor sample. The absence of alterations in genes with low coverage should be interpreted carefully in the context of the patient's diagnosis, with consideration for retesting. Variants are identified through aligning the patient's DNA sequence to the human genome reference sequence, version hg19 (chr1-22,X,Y).

Tumor mutational burden (TMB) measures the quantity of somatic SNVs and indels, of any pathogenicity, including benign, carried in a tumor as the number of protein-altering mutations per million coding base pairs. TMB is calculated at the time of initial report delivery. Accordingly, the TMB calculation is based upon (a) both the tumor and normal sample if Tempus had analyzed both at the time of the initial report, or (b) the tumor sample only if no normal sample had been analyzed at the time of the initial report. Please note that tumor only calculations are not updated or amended even if a normal sample is subsequently analyzed. Studies have shown that tumors with higher TMB have an increased likelihood of response to immunotherapy [1, 2].

Microsatellite instability (MSI) refers to hypermutability caused by genetic or acquired defects in the DNA mismatch repair pathway. MSI status is divided into MSI-high (MSI-H) tumors, which have changes in microsatellite repeat lengths due to defective DNA mismatch repair activity. Microsatellite stable (MSS) tumors do not have detectable defects in DNA mismatch repair. Microsatellite equivocal (MSE) tumors have an intermediate phenotype which cannot be clearly classified as MSI-H or MSS based on the statistical cutoff used to define those categories. If MSI status will affect clinical management, immunohistochemical staining for DNA mismatch repair proteins, or another method of ascertaining MSI status, is recommended.

1. Nivolumab plus ipilimumab in Lung Cancer with a High Tumor Mutational Burden. <https://www.ncbi.nlm.nih.gov/pubmed/29658845>

2. Cancer Immunology: Mutational landscape determines sensitivity to PD-1 blockade in non-small cell lung cancer. <https://www.ncbi.nlm.nih.gov/pubmed/25765070>

Gene List

A-B

TEMPUS

Electronically Signed By:
Rongqin Ren, MD, PhD

CLIA Number:
14D2114007

Date Signed/Reported:
06/05/2024

Laboratory Medical Director:
Brett Mahon, MD, FACP

Tempus ID #:
TL-24-AU2MEAY

Pipeline Version:
5.1.0

4/8

Tempus AI, Inc. • 600 West Chicago Avenue, Ste 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com

ABCB1, ABCG3, ABL1, ABL2, ABRAXAS1, ACTA2, ACVR1 (ALK2), ACVR1B, AGO1, AJUBA, AKT1, AKT2, AKT3, ALK, AMER1, APC, APLNR, APOB, AR, ARAF, ARHGAP26, ARHGAP35, ARID1A, ARID1B, ARID2, ARID5B, ASNS, ASPSCR1, ASXL1, ATIC, ATM, ATP7B, ATR, ATRX, AURKA, AURKB, AXIN1, AXIN2, AXL, BLM, BAPI, BARD1, BCL10, BCL11B, BCL2, BCL2L1, BCL6, BCL7A, BCLAF1, BCOR, BCORL1, BCR, BIRC3, BLM, BMPR1A, BRAF, BRCA1, BRCA2, BRD4, BRD9, BTG1, C1R, C1RL

[illegible]

七、十

[illegible]

5.14

GPR9, GARRA6, GALNT12, GATA1, GATA2, GATA3, GATA4, GATA6, GENT1, GLI1, GLI2, GNA11, GNA13, GNAQ, GNAS, GPC3, GPS2, GREM1, GRIN2A, GRM2, GSTP1, H19, H3F3A, H3F3, HAVCR2, HDAC1, HDAC2, HDAC4, HGF, HIF1A, HIST1H1E, HIST1H3B, HIST1H4E, HLA-A, HLA-B, HLA-C, HLA-DMA, HLA-DMB, HLA-DOA, HLA-DOB, HLA-DPA1, HLA-DPB1, HLA-DPB2, HLA-DQA1, HLA-DQA2, HLA-DQB1, HLA-DQB2, HLA-DRA, HLA-DRB1, HLA-DRB5, HLA-DRB6, HLA-E, HLA-F, HLA-G, HNTF1, HNF1B, HMOX1A1, HMOX1B3, HRAS, HSD11B2, HSD3B1, HSD3B2, HSP90AA1, HSPH1

I-K

IDH1, IDH2, IDO1, IFIT1, IFIT2, IFIT3, IFNAR1, IFNAR2, IFNGR1, IFNGR2, IFNL3, IKKBE, IKZF1, IL10RA, IL15, IL2RA, IL6R, IL7R, ING1, INPP4B, IRF1, IRF2, IRF4, IRS2, ITPKB, JAK1, JAK2, JAK3, JUN, KAT6A, KDMA, KDM5C, KDM5D, KDM6A, KDE, KEAP1, KEL, KIF1B, KIT, KLF4, KLHL6, KLLN, KMT2A, KMT2B,
KMT2C, KMT2D, KMT2E

[illegible]

4. N1

L2HGDI, LAG3, LAT51, LCK, LCL, LEFT, LMNA, LMO1, LRP1B, LYN, LZTR1, MAD2L2, MAF, MAFB, MAGI2, MALT1, MAP2K1, MAP2K2, MAP2K4, MAP3K1, MAP3K7, MAPK1, MAPK, MC1R, MCTP1, MDM2, MDM4, MED12, MEF2B, MEN1, ME, MGMT, MIB1, MITF, MKI67, MLH1, MLH3, MLT3, MN1, MPL, MRE11, MSA1, MSK2, MSK3, MSK4, MTA1, MTHFD2, MTHFR, MTRR, MTRR, MUTHV, MYB, MYC, MYCL, MYCN, MYD88, MYH11

NRN, NCOR1, NCOR2, NF1, NF2, NFEBL2, NFKBIA, NHP2, NKX2-1, NOP10, NOTCH1, NOTCH2, NOTCH3, NOTCH4, NPM1, NQO1, NRAS, NRG1, NSD1, NSD2, NTSC2, NTHL1, NTRK1, NTRK2, NTRK3, NUDT15, NUP98, OLIG2

PC

P2RX8, PAK1, PALB2, FALD1, PAX3, PAX5, PAX7, PAX8, PBRM1, PCBP1, PDCD1, PDCD1LG2, PDGFRFA, PDGFRA, PDGFRB, PDK1, PHF6, PHGDH, PHLP1, PHLP2, PHOX2B, PIAS4, PIK3C2B, PIK3CA, PIK3CB, PIK3CD, PIK3CG, PIK3R1, PIK3R2, PIM1, PLCG1, PLCG2, PML, PMS1, PMS2, POLD1, POLE, POLH, POLQ, POT1, PTEN, PRKAA, PRKAB, PRKAC, PRKAD, PRKE, PRKF, PRKG1, PRKG2, PRKG3, PRKG4, PRKG5, PRKG6, PRKG7, PRKG8, PRKG9, PRKG10, PRKG11, PRKG12, PRKG13, PRKG14, PRKG15, PRKG16, PRKG17, PRKG18, PRKG19, PRKG20, PRKG21, PRKG22, PRKG23, PRKG24, PRKG25, PRKG26, PRKG27, PRKG28, PRKG29, PRKG30, PRKG31, PRKG32, PRKG33, PRKG34, PRKG35, PRKG36, PRKG37, PRKG38, PRKG39, PRKG40, PRKG41, PRKG42, PRKG43, PRKG44, PRKG45, PRKG46, PRKG47, PRKG48, PRKG49, PRKG50, PRKG51, PRKG52, PRKG53, PRKG54, PRKG55, PRKG56, PRKG57, PRKG58, PRKG59, PRKG60, PRKG61, PRKG62, PRKG63, PRKG64, PRKG65, PRKG66, PRKG67, PRKG68, PRKG69, PRKG70, PRKG71, PRKG72, PRKG73, PRKG74, PRKG75, PRKG76, PRKG77, PRKG78, PRKG79, PRKG80, PRKG81, PRKG82, PRKG83, PRKG84, PRKG85, PRKG86, PRKG87, PRKG88, PRKG89, PRKG90, PRKG91, PRKG92, PRKG93, PRKG94, PRKG95, PRKG96, PRKG97, PRKG98, PRKG99, PRKG100, PRKG101, PRKG102, PRKG103, PRKG104, PRKG105, PRKG106, PRKG107, PRKG108, PRKG109, PRKG110, PRKG111, PRKG112, PRKG113, PRKG114, PRKG115, PRKG116, PRKG117, PRKG118, PRKG119, PRKG120, PRKG121, PRKG122, PRKG123, PRKG124, PRKG125, PRKG126, PRKG127, PRKG128, PRKG129, PRKG130, PRKG131, PRKG132, PRKG133, PRKG134, PRKG135, PRKG136, PRKG137, PRKG138, PRKG139, PRKG140, PRKG141, PRKG142, PRKG143, PRKG144, PRKG145, PRKG146, PRKG147, PRKG148, PRKG149, PRKG150, PRKG151, PRKG152, PRKG153, PRKG154, PRKG155, PRKG156, PRKG157, PRKG158, PRKG159, PRKG160, PRKG161, PRKG162, PRKG163, PRKG164, PRKG165, PRKG166, PRKG167, PRKG168, PRKG169, PRKG170, PRKG171, PRKG172, PRKG173, PRKG174, PRKG175, PRKG176, PRKG177, PRKG178, PRKG179, PRKG180, PRKG181, PRKG182, PRKG183, PRKG184, PRKG185, PRKG186, PRKG187, PRKG188, PRKG189, PRKG190, PRKG191, PRKG192, PRKG193, PRKG194, PRKG195, PRKG196, PRKG197, PRKG198, PRKG199, PRKG200, PRKG201, PRKG202, PRKG203, PRKG204, PRKG205, PRKG206, PRKG207, PRKG208, PRKG209, PRKG210, PRKG211, PRKG212, PRKG213, PRKG214, PRKG215, PRKG216, PRKG217, PRKG218, PRKG219, PRKG220, PRKG221, PRKG222, PRKG223, PRKG224, PRKG225, PRKG226, PRKG227, PRKG228, PRKG229, PRKG230, PRKG231, PRKG232, PRKG233, PRKG234, PRKG235, PRKG236, PRKG237, PRKG238, PRKG239, PRKG240, PRKG241, PRKG242, PRKG243, PRKG244, PRKG245, PRKG246, PRKG247, PRKG248, PRKG249, PRKG250, PRKG251, PRKG252, PRKG253, PRKG254, PRKG255, PRKG256, PRKG257, PRKG258, PRKG259, PRKG260, PRKG261, PRKG262, PRKG263, PRKG264, PRKG265, PRKG266, PRKG267, PRKG268, PRKG269, PRKG270, PRKG271, PRKG272, PRKG273, PRKG274, PRKG275, PRKG276, PRKG277, PRKG278, PRKG279, PRKG280, PRKG281, PRKG282, PRKG283, PRKG284, PRKG285, PRKG286, PRKG287, PRKG288, PRKG289, PRKG290, PRKG291, PRKG292, PRKG293, PRKG294, PRKG295, PRKG296, PRKG297, PRKG298, PRKG299, PRKG300, PRKG301, PRKG302, PRKG303, PRKG304, PRKG305, PRKG306, PRKG307, PRKG308, PRKG309, PRKG310, PRKG311, PRKG312, PRKG313, PRKG314, PRKG315, PRKG316, PRKG317, PRKG318, PRKG319, PRKG320, PRKG321, PRKG322, PRKG323, PRKG324, PRKG325, PRKG326, PRKG327, PRKG328, PRKG329, PRKG330, PRKG331, PRKG332, PRKG333, PRKG334, PRKG335, PRKG336, PRKG337, PRKG338, PRKG339, PRKG340, PRKG341, PRKG342, PRKG343, PRKG344, PRKG345, PRKG346, PRKG347, PRKG348, PRKG349, PRKG350, PRKG351, PRKG352, PRKG353, PRKG354, PRKG355, PRKG356, PRKG357, PRKG358, PRKG359, PRKG360, PRKG361, PRKG362, PRKG363, PRKG364, PRKG365, PRKG366, PRKG367, PRKG368, PRKG369, PRKG370, PRKG371, PRKG372, PRKG373, PRKG374, PRKG375, PRKG376, PRKG377, PRKG378, PRKG379, PRKG380, PRKG381, PRKG382, PRKG383, PRKG384, PRKG385, PRKG386, PRKG387, PRKG388, PRKG389, PRKG390, PRKG391, PRKG392, PRKG393, PRKG394, PRKG395, PRKG396, PRKG397, PRKG398, PRKG399, PRKG400, PRKG401, PRKG402, PRKG403, PRKG404, PRKG405, PRKG406, PRKG407, PRKG408, PRKG409, PRKG410, PRKG411, PRKG412, PRKG413, PRKG414, PRKG415, PRKG416, PRKG417, PRKG418, PRKG419, PRKG420, PRKG421, PRKG422, PRKG423, PRKG424, PRKG425, PRKG426, PRKG427, PRKG428, PRKG429, PRKG430, PRKG431, PRKG432, PRKG433, PRKG434, PRKG435, PRKG436, PRKG437, PRKG438, PRKG439, PRKG440, PRKG441, PRKG442, PRKG443, PRKG444, PRKG445, PRKG446, PRKG447, PRKG448, PRKG449, PRKG450, PRKG451, PRKG452, PRKG453, PRKG454, PRKG455, PRKG456, PRKG457, PRKG458, PRKG459, PRKG460, PRKG461, PRKG462, PRKG463, PRKG464, PRKG465, PRKG466, PRKG467, PRKG468, PRKG469, PRKG470, PRKG471, PRKG472, PRKG473, PRKG474, PRKG475, PRKG476, PRKG477, PRKG478, PRKG479, PRKG480, PRKG481, PRKG482, PRKG483, PRKG484, PRKG485, PRKG486, PRKG487, PRKG488, PRKG489, PRKG490, PRKG491, PRKG492, PRKG493, PRKG494, PRKG495, PRKG496, PRKG497, PRKG498, PRKG499, PRKG500, PRKG501, PRKG502, PRKG503, PRKG504, PRKG505, PRKG506, PRKG507, PRKG508, PRKG509, PRKG510, PRKG511, PRKG512, PRKG513, PRKG514, PRKG515, PRKG516, PRKG517, PRKG518, PRKG519, PRKG520, PRKG521, PRKG522, PRKG523, PRKG524, PRKG525, PRKG526, PRKG527, PRKG528, PRKG529, PRKG530, PRKG531, PRKG532, PRKG533, PRKG534, PRKG535, PRKG536, PRKG537, PRKG538, PRKG539, PRKG540, PRKG541, PRKG542, PRKG543, PRKG544, PRKG545, PRKG546, PRKG547, PRKG548, PRKG549, PRKG550, PRKG551, PRKG552, PRKG553, PRKG554, PRKG555, PRKG556, PRKG557, PRKG558, PRKG559, PRKG560, PRKG561, PRKG562, PRKG563, PRKG564, PRKG565, PRKG566, PRKG567, PRKG568, PRKG569, PRKG570, PRKG571, PRKG572, PRKG573, PRKG574, PRKG575, PRKG576, PRKG577, PRKG578, PRKG579, PRKG580, PRKG581, PRKG582, PRKG583, PRKG584, PRKG585, PRKG586, PRKG587, PRKG588, PRKG589, PRKG590, PRKG591, PRKG592, PRKG593, PRKG594, PRKG595, PRKG596, PRKG597, PRKG598, PRKG599, PRKG600, PRKG601, PRKG602, PRKG603, PRKG604, PRKG605, PRKG606, PRKG607, PRKG608, PRKG609, PRKG610, PRKG611, PRKG612, PRKG613, PRKG614, PRKG615, PRKG616, PRKG617, PRKG618, PRKG619, PRKG620, PRKG621, PRKG622, PRKG623, PRKG624, PRKG625, PRKG626, PRKG627, PRKG628, PRKG629, PRKG630, PRKG631, PRKG632, PRKG633, PRKG634, PRKG635, PRKG636, PRKG637, PRKG638, PRKG639, PRKG640, PRKG641, PRKG642, PRKG643, PRKG644, PRKG645, PRKG646, PRKG647, PRKG648, PRKG649, PRKG650, PRKG651, PRKG652, PRKG653, PRKG654, PRKG655, PRKG656, PRKG657, PRKG658, PRKG659, PRKG660, PRKG661, PRKG662, PRKG663, PRKG664, PRKG665, PRKG666, PRKG667,

R-S

RAC1, RAD21, RAD50, RAD51, RAD51AP1, RAD51C, RAD51D, RAD54A, RAD54B, RAF1, RANBP2, RARA, RAS1, RB1, RBM10, RECQL4, REL, RHBF, RHOA, RICTOR, RPLP0, RPLP2, RPLP3, RPLP4, RPLP5, RPLP6, RPS15A, RPS18, RPS27A, RPS29, RPTOR, RPLR1, RSF1, RUVBL1, RXRA, SCHEC3, SDHA, SDHAF2, SDBH, SDHC, SDHD, SETD3, JEMAD3, SETPB1, SETPD1, SF3B1, SGK1, SH2B3, SH4, SLG26A3, SLG47A2, SLXCAR1, SLIT1, SLK, SLMO2, SMAD2, SMAD3, SMAD4, SMARCA1, SMARCA4, SMARCB1, SMARCE1, SMCA1, SMCA3, SMO, SOCS1, SOD2, SOX10, SOX2, SOX9, SPEN, SPINK1, SPOD, SPRED1, SRC, SRSF2, STAG2, STAT3, STAT4, STAT5A, STAT5B, STAT6, STRT1, SUZ12, SUV39H1, SVW, SYCE1

1

TAFL, TANC1, TAP1, TAP2, TARSP2, TBC1D12, TBL1XR1, TBX3, TCF3, TCF7L2, TCL1A, TERT, TET2, TFE3, TFEB, TFEC, TGFBR1, TGFBR2, TIGIT, TMEM127, TMEM138, TMEM259, TNP1, TNPAP1, TNFRSF14, TNFRSF17, TNFRSF9, TOP1, TOP2A, TP53, TP63, TPM1, TPMT, TRAF3, TRAF7, TSC1, TSC2, TSHR, TUSC3, TYMS, U2AF1, UBE2T, UGT1A1, UGT1A9, UMPF

Y.

Alan Linquist

Diagnosis
Dedifferentiated chondrosarcomaAccession No.
TL-24-BV3UXZAM

xR

Date of Birth
06/07/1955Sex
MalePhysician
John CharlsonInstitution
MCW - Froedtert Hospital Cancer
Center
E2304021TEMPUS | xR
TranscriptomeTumor specimen:
Soft tissue, chest wall
Wisconsin Diagnostics Laboratories -
Pathology 15-23-18596, D1
Collected 8/1/2023
Received 5/28/2024
Tumor Percentage: 80%
(post microdissection)THIS REPORT IS BEING ISSUED TO REPORT THE RESULTS OF GENE
REARRANGEMENT AND ALTERED SPLICING ANALYSIS FROM RNA
SEQUENCING.No gene rearrangements nor reportable altered splicing events were
identified from RNA sequencing.

GENOMIC VARIANTS

Potentially Actionable / Biologically Relevant

No reportable pathogenic variants were found.

TREATMENT IMPLICATIONS

No reportable treatment options found.

TEMPUS

Lab Version
14D2114007Date Issued
06/06/2024Laboratory Medical Director
Brett Mahon, MD, FCAPTempus ID #
TL-24-BV3UXZAMPipeline Version
3.6.1 1/3

Tempus, AT Inc. • 400 West Chicago Avenue, Ste 410 • Chicago, IL • 60654 • tempus.com • support@tempus.com

Alan Linquist | TL-24-BV3UXZAM

Assay Description

The Tempus xR test is a whole-transcriptome genomic sequence analysis panel for solid tumor and hematological malignancy samples that reports clinically relevant chromosomal rearrangements arising from the genetic rearrangement of more than 100 targeted genes, as well as altered splicing for MET exon 14 and EGFRvIII in an unbiased and comprehensive manner. Chromosomal rearrangements and altered splicing events are detected by robust capture next-generation sequencing (NGS) using custom designed IDT probes. Test results are intended to provide tumor molecular information that may be used by clinicians to help inform clinical management when patients are seeking further cancer treatment. This assay has 100% sensitivity for targeted chromosomal rearrangements, 97% sensitivity for untargeted chromosomal rearrangements, and 95.5% sensitivity for altered splicing at 20% tumor purity.

The clinical summary (first page of the report) shows potentially actionable and biologically relevant chromosomal rearrangements and altered splicing variants. RNA chromosomal rearrangement and altered splicing event analysis is performed without a matched normal sample and so results do not distinguish whether reported chromosomal rearrangements or altered splicing events are germline or somatic. **Potentially Actionable** alterations have an associated therapeutic, diagnostic, and/or prognostic indication, based on evidence from US-based clinical guidelines and medical literature, curated in the Tempus knowledge database. **Biologically Relevant** alterations may have functional significance or have been observed in the medical literature, but do not have relevant therapeutic, diagnostic, or prognostic evidence curated in the Tempus knowledge database. Benign alterations of non-cancerous cells are identified by aligning the patient's RNA sequence to the human genome reference sequence version hg19 (GRCh37).

The xR assay uses IDT xGen Exome Research Panel v2 probe set as backbone, which consists of >415K individually synthesized probes and spans 34 Mb target region (19.4% of the human genome). Additional Tempus-specific custom spike-in probes are added to enhance target region detection. When whole transcriptome RNA-Seq is performed, expressed fusion transcripts from chromosomal rearrangements specifically targeted by the assay will be detected. In addition, expressed fusion transcripts from rearranged genes not specifically targeted by the assay may also be detected. A list of targeted fusion genes and targeted altered splicing events can be made available upon request. The bioinformatics chromosomal rearrangement transcript detection pipeline identifies and analyzes the positions of breakpoint-spanning reads and split paired-end reads. Non-canonical chromosomal rearrangement transcripts may be reported at the discretion of the pathologist. The altered splicing bioinformatics pipeline analyzes the positions of split paired-end reads spanning annotated and novel exon-exon junctions.

Tempus Disclaimer

This analysis of nucleic acids by next-generation sequencing (NGS) can be affected by multiple factors including formalin-fixation, degrading RNA quality, and low tumor purity limiting sensitivity. Additionally, the chance of detecting genetic alterations may be reduced in regions of the genome which are structurally difficult to sequence, in homologous genes, or due to low complexity regions that are prone to sequencing error.

Genetic alterations are defined as clinically significant based on peer-reviewed published literature and other authoritative sources such as NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). These references are not comprehensive, therefore clinically unknown findings may occur. Tempus does not notify physicians or patients of updated variant classifications.

These test results and information contained within the report are current as of the report date. Tempus will not update reports or send notification regarding reclassification of genomic alterations. This test was developed and its performance characteristics were determined by Tempus AI, Inc. It has not been cleared or approved by the US Food and Drug Administration. The laboratory is CLIA certified to perform high-complexity testing. Any decisions related to patient care and treatment choices should be based on the independent judgment of the treating physician and should take into account all information related to the patient, including without limitation, the patient and family history, direct physical examination, and other tests. Tempus is not liable for medical judgment with regards to diagnosis, prognosis, or treatment in connection with the test results.

Microdissection may have been performed at Tempus AI, Inc., either at 25 Alexandria Way, Ste 300, Durham, NC 27703 (CLIA# 34D2228314), or 600 West Chicago Ave, Ste 510, Chicago, IL 60654 (CLIA# 14D2114007). Bioinformatics processing was performed at Tempus AI, Inc. (CLIA# 14D2114007), or 600 West Chicago Ave, Ste 510, Chicago, IL 60654.

Dates and times are represented in the coordinated universal time zone (UTC) unless otherwise specified. However, dates that are provided to Tempus without a timestamp (e.g., sample collection date) are listed as provided.

TEMPUS

CLIA Number
14D2114007

Date Issued
06/06/2024

Laboratory Medical Director
Brett Mahon, MD, FCAP

Tempus ID #
TL-24-BV3UXZAM

Pipeline Version
3.6.1 2/3

Tempus AI, Inc • 600 West Chicago Avenue, Ste 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com

Alan Linquist | TL-24-BV3UXZAM

Tempus Disclaimer (continued)

Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) Acute Lymphoblastic Leukemia, Version: 4.2023, Acute Myeloid Leukemia, Version: 1.2024, Ampullary Adenocarcinoma, Version: 1.2024, Anal Carcinoma, Version: 1.2024, Basal Cell Skin Cancer, Version: 3.2024, B-Cell Lymphomas, Version: 1.2024, Biliary Tract Cancers, Version: 3.2023, Bladder Cancer, Version: 1.2024, Bone Cancer, Version: 1.2024, Breast Cancer, Version: 1.2024, Castleman Disease, Version: 1.2024, Central Nervous System Cancers, Version: 1.2023, Cervical Cancer, Version: 2.2024, Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma, Version: 1.2024, Chronic Myeloid Leukemia, Version: 2.2024, Colon Cancer, Version: 1.2024, Dermatofibrosarcoma Protuberans, Version: 1.2024, Esophageal and Esophagogastric Junction Cancers, Version: 1.2024, Gastric Cancer, Version: 1.2024, Gastrointestinal Stromal Tumors, Version: 1.2024, Gestational Trophoblastic Neoplasia, Version: 1.2024, Hairy Cell Leukemia, Version: 1.2024, Head and Neck Cancers, Version: 3.2024, Hepatobiliary Cancers, Version: 1.2023, Hepatocellular Carcinoma, Version: 2.2023, Histiocytic Neoplasms, Version: 1.2023, Hodgkin Lymphoma, Version: 2.2024, Kaposi Sarcoma, Version: 1.2024, Kidney Cancer, Version: 2.2023, Melanoma, Cutaneous, Version: 1.2024, Melanoma: Uveal, Version: 1.2023, Merkel Cell Carcinoma, Version: 1.2024, Mesothelioma: Peritoneal, Version: 1.2024, Mesothelioma: Pleural, Version: 1.2024, Multiple Myeloma, Version: 3.2024, Myelodysplastic Syndromes, Version: 1.2024, Myeloid/Lymphoid Neoplasms with Eosinophilia and Tyrosine Kinase Gene Fusions, Version: 1.2024, Myeloproliferative Neoplasms, Version: 1.2024, Neuroblastoma, Version: 1.2024, Neuroendocrine and Adrenal Tumors, Version: 1.2023, Non-Small Cell Lung Cancer, Version: 2.2024, Occult Primary, Version: 1.2024, Ovarian Cancer/Ectopic Tube Cancer/Primary Peritoneal Cancer, Version: 1.2024, Pancreatic Adenocarcinoma, Version: 1.2024, Pediatric Acute Lymphoblastic Leukemia, Version: 4.2024, Pediatric Aggressive Mature B-Cell Lymphomas, Version: 1.2023, Pediatric Central Nervous System Cancers, Version: 1.2024, Pediatric Hodgkin Lymphoma, Version: 2.2023, Penile Cancer, Version: 1.2024, Primary Cutaneous Lymphomas, Version: 1.2024, Prostate Cancer, Version: 3.2024, Rectal Cancer, Version: 1.2024, Small Bowel Adenocarcinoma, Version: 2.2024, Small Cell Lung Cancer, Version: 2.2024, Soft Tissue Sarcoma, Version: 3.2023, Squamous Cell Skin Cancer, Version: 1.2024, Systemic Light Chain Amyloidosis, Version: 2.2024, Systemic Mastocytosis, Version: 1.2024, T-Cell Lymphomas, Version: 1.2024, Testicular Cancer, Version: 1.2023, Thymomas and Thymic Carcinomas, Version: 1.2024, Thyroid Carcinoma, Version: 1.2024, Uterine Neoplasms, Version: 2.2024, Vulvar Cancer, Version: 3.2024, Waldenström Macroglobulinemia / Lymphoplasmacytic Lymphoma, Version: 2.2024, Wilms Tumor (Nephroblastoma), Version: 1.2023, Version: 1.2023 & National Comprehensive Cancer Network, Inc. 2024. All rights reserved. Accessed March 7, 2024. To view the most recent and complete version of this guideline, go online to National Comprehensive Cancer Network - Home. The NCCN Guidelines® and other content provided by NCCN are works in progress that may be refined as often as new significant data becomes available. They are statements of consensus of its authors regarding their view of currently accepted approaches to treatment. Any clinician seeking to apply or consult any NCCN Guidelines® or other NCCN Content is expected to use independent medical judgment in the context of individual clinical circumstances to determine any patient's care or treatment. NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way. Therapeutic options are not applicable in all disease settings.

The OncorB® precision oncology knowledge base was made available under license from Memorial Sloan Kettering Cancer Center. See terms [here](#).

© 2024 Tempus Labs, Inc.
All rights reserved.

TEMPUS

CLIA Number:
14D2114007

Date Issued:
06/06/2024

Laboratory Medical Director
Brett Mahon, MD, FACP

Tempus ID #
TL-24-BV3UXZAM

Pipeline Version
3.6.1 3/3

Tempus, AL-001-806 West Chicago Avenue, Ste 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com

FOR RESEARCH USE ONLY

Date of Birth
06/07/1955Sex
MalePhysician
John CharlsonInstitution
MCW - Froedtert Hospital Cancer
Center
E2304021

TEMPUS | RNA

Transcriptome

Tumor specimen:
Soft tissue, chest wall
Wisconsin Diagnostics Laboratories -
Pathology 1-S-23-18596, D1
Collected 8/1/2023
Received 5/28/2024
Tumor Percentage: 80%
(post microdissection)

Molecular Profile

The Tempus RNA-Seq Report provides information about mRNA expression levels in the patient's tumor to provide insights into the patient's tumor biology. The information is provided for informational purposes only and is not intended for use in clinical decision making outside the research setting.

EXPRESSION DETAILS

AKT2	Overexpressed
MAP2K2	Overexpressed
CDKN2B	Underexpressed
RET	Overexpressed
MAP2K1	Overexpressed

IMMUNOTHERAPY MARKERS

Tempus Disclaimer

This report was developed and its performance characteristics determined by Tempus AI, Inc. It has not been cleared or approved by the US Food and Drug Administration. The laboratory is CLIA certified to perform high-complexity testing. The performance characteristics of the molecular profile have been established, but the clinical meaning of the data has not been established. The information is provided for informational purposes only and is not intended for use in clinical decision making outside the research setting. Any diagnosis, counseling, or treatment determination should be made by a qualified health-care professional in conjunction with other individual patient health information, including clinical presentation and the results of other diagnostic tests, as may be available.

Dates and times are represented in the coordinated universal time zone (UTC) unless otherwise specified. However, dates that are provided to Tempus without a timestamp (e.g., sample collection date) are listed as provided.

CLINICAL TRIALS

TEMPUS is testing the use of liquid and solid tumor biopsy specimens to identify potential drug targets and biomarkers for clinical trials. This report is generated from a liquid biopsy specimen collected from a patient with dedifferentiated chondrosarcoma.

Tempus AI
1000 Wisconsin Ave, 5th Fl
Chicago, IL 60601-2000
USA
Phone: 312.555.1234
Email: support@tempus.com

TEMPUS

Date issued
06/06/2024Tempus ID #
TL-24-Z51683CGPipeline Version
3.6.1 1/1

Tempus AI, Inc. • 600 West Chicago Avenue, Ste 510 • Chicago, IL • 60654 • tempus.com • support@tempus.com